

Feng Chia University Course Syllabus (Full Version)

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IND3411 PYTHON PROGRAMMING & APPLICATIONS 3 Credits

Course Description

The course introduces the essential Python programming language to learn the controlling skill for the robotics sensors and vision. The contents of the course include the basic programming skill of coding variables for data storage, operation of data, the control flow of behavior change, and create function. In addition, the basic concept of controlling robot sensors and vision system will be introduced throughout the course.

Pre-course :

Class Information

Semester : 112-1

Course Period : Full semester

Course Selection Code : 0002

Course Time/Course Room : (Fri)02-04/JCB611

Teaching Hours : 3

Language of Instruction : English

Course Delivery Mode : Face to Face Course

Do you use M365 Teams for class with video (recording) tools : Yes

EMI Course : Yes

Class : General Program, School of Architecture

Course Property(Credits) : Elective(3)

Instructor / Teaching Assistant Information

Instructor

Name	Office Hours/Room	E-mail	Extension
Yee-Siang Gan	(三)03 忠807B,(三)07-08 忠807B,(三)09 忠807B	ysgan@o365.fcu.edu.tw	3306

Teaching Assistant

Name	Office Hours/Room	E-mail	Extension
WU, YI-LIANG		m1202354@o365.fcu.edu.tw	

Course Objectives

- 1.Able to understand the logic of basic programming and limitations.
- 2.Able to apply python programming to write robot' s algorithm and interpretation.
- 3.Able to analyze the condition of robot' s path planning and evaluate the performance of robot' s algorithm

Textbook & Course Materials

Textbook

Python Crash Course, 3rd Edition: A Hands-On, Project-Based Introduction to Programming,Eric Matthes

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Reading

Python Cookbook, Third Edition3rd EditionbyDavid Beazley(Author),Brian Jones(Author),Publisher: O'Reilly Media

Learning Tools

iTools

Order	Software tools	Explanation
1	Python; spyder	recommended window 10

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Course Schedule and Outlines

Week/Unit	Topic & Readings	Teaching and learning activities
Weeks 1-2	Course Introduction and Basic Syntax -Course objectives and overview Features and applications of the Python programming language Setting up the Python development environment - Basic syntax: variables, data types, operators	
Weeks 3-4	Conditional Statements and Control Structures -Conditional statements: if, elif, else- Logical and comparison operators- Loops: for loop, while loop	
Weeks 5	Functions -Function definition and invocation-Function parameters and return values-Built-in functions and user-defined functions	
Weeks 6	Modules -Module usage and import	
Weeks 7	Midterm exam	
Week8	File Handling&Error Handling -Reading and writing files-Exception handling mechanisms -try-except-finally syntax-Common errors and troubleshooting	
Weeks 9-10	Object-Oriented Programming (OOP) -OOP concepts and features-Creating and using classes and objects-Inheritance, polymorphism, and encapsulation-Object-oriented design principles	
Weeks 11	Robot Motion Simulation -Creating robot characters-Handling user input controls-Implementing basic motion simulations	
Week12	Web Applications	
Week13	Implementation of small-scale Python applications	
Week14	Pygame Application I	
Week15	Pygame Application II	
Week16	Final Presentation	

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Grading Policy

Assignments & Assessments

Assignments	Calculating grades based on weight ratio	Explanation
General impression	5%	Assessing students' engagement, questioning, discussions, and collaborative skills in class.
Attendances	10%	Randomly selected at least 10 times.
Homework	15%	Regular assignments given during class.
Midterm examination	30%	Comprehensive exam covering foundational to advanced framework knowledge
Final presentation	40%	Students complete a programming project demonstrating practical Python application in real-world scenarios

Description of scoring rules

Assignments and Programming Projects (55%): Students are required to complete a series of programming assignments covering various topics and difficulty levels. At the end of the course, they will submit a programming project demonstrating practical Python application in real-world scenarios. Midterm Exam (30%): A midterm exam will be conducted to assess understanding of fundamental concepts, syntax, and skills in Python. Participation and Classroom Performance (10%): Assessing students' engagement, questioning, discussions, and collaborative skills in class. Extra Credit (optional, up to 10%): Encouraging participation in extracurricular activities such as programming competitions, open-source projects, coding communities, and submission of relevant proof materials.

Academic Integrity Statements

In the academic realm, academic integrity is of utmost importance. We expect students to demonstrate independent thinking and originality in their learning assessments and to maintain respect and collaboration in classroom interactions. Sustaining academic integrity beyond the course is foundational to establishing a professional reputation. Violations of academic integrity will result in severe consequences, including grade deductions, academic warnings, or more stringent penalties. Let's collectively uphold academic ethics to ensure a fair and honest learning environment.

Course Policies

In this course, we place a strong emphasis on good classroom etiquette and online decorum. For in-person classes, please arrive on time, stay focused, and actively participate while respecting your peers. Promptly asking questions and sharing insights is encouraged. Maintaining academic integrity by avoiding plagiarism and unfair competition is essential in all contexts. This approach fosters an environment of mutual respect and support. Non-compliance with these norms may result in grade deductions or appropriate penalties.

Course Friendly Reminder

We aspire to create a classroom environment characterized by mutual respect and support. Remember the importance of punctuality to ensure your full engagement in the course material. Don't hesitate to raise questions or share your viewpoints during class discussions; this enriches our discourse. Moreover, feel free to seek assistance whenever needed, as we are here to support your learning journey. Meanwhile, please maintain academic integrity by refraining from improper citations and copying. If you require extra learning resources, we will offer relevant recommendations to enhance your mastery of the course content.

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Correlation of Course Objectives, Theme of Learning Levels, Teaching Strategies and Assessment Methods for Basic/Core Learning Outcomes

Learning Outcomes	Course Objectives	Learning Levels	Teaching Strategies	Assessment Method
B.To obtain inter-disciplinary integration ability in terms of planning, project-based research, demonstration of learning, communication and coordination.	Able to understand the logic of basic programming and limitations.	1	Classroom instruction 、 Extensive reading 、 Inquiry-based Learning, IBL 、 Problem-based Learning, PBL	Midterm examination 、 Homework 、 Final presentation 、 General impression 、 Participation 、 Attendances
E.To obtain innovative and independent thinking, under a macro and global vision.	Able to apply python programming to write robot' s algorithm and interpretation.	3	Classroom instruction 、 Seminar 、 Written report 、 Inquiry-based Learning, IBL 、 Problem-based Learning, PBL	Midterm examination 、 Homework 、 Final presentation 、 General impression 、 Participation 、 Attendances
E.To obtain innovative and independent thinking, under a macro and global vision.	Able to analyze the condition of robot' s path planning and evaluate the performance of robot' s algorithm	5	Seminar 、 Written report 、 Problem-based Learning, PBL 、 Project-based Learning, PjBL	Midterm examination 、 Homework 、 Final presentation 、 General impression 、 Roll call

The SDGs' Issues or Contents are Immersed in Teaching and Learning Activities

SDGs topic or content	Correlation
Economic Growth-Industry, Innovation and Infrastructure	Indirect correlation